

ILE BICQUETTE, QC, Canada

WMO: 713850

Lat: 48.415N

Lon: 68.893W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.6	-25.8	0.4	-20.6	-23.4	0.5	-18.5	20.6	-1.9	18.5	-6.9	9.0	290	0.903

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.9	(c) 19.2	(d) 15.4	(e) 17.8	(f) 14.6	(g) 16.6	(h) 13.9	(i) 16.3	(j) 18.1	(k) 15.3	(l) 16.9	(m) 14.4	(n) 16.0	(o) 4.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	11.0	17.1	14.4	10.3	16.1	13.5	9.7	15.3	45.7	18.2	42.8	17.0	40.3	16.1	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.5	(b) 15.1	(c) 13.5	(e) -23.3	(f) 23.5	(g) 2.3	(h) 1.5	(i) -25.0	(j) 24.6	(k) -26.3	(l) 25.5	(m) -27.6	(n) 26.3	(o) -29.2	(p) 27.4		
(4) 16.5	(5) 15.1	(6) 13.5	(7) -23.6	(8) 19.1	(9) 1.5	(10) 0.8	(11) -24.6	(12) 19.6	(13) -25.5	(14) 20.1	(15) -26.4	(16) 20.5	(17) -27.4	(18) 21.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-9.1	-7.8	-3.1	2.5	7.3	11.2	13.3	12.8	10.1	6.0	1.3	-4.9	
	DBStd	8.44	5.66	5.17	4.40	2.78	2.37	2.30	2.04	1.96	2.15	2.40	3.41	4.74	
	HDD10.0	2701	591	498	406	224	90	14	1	2	25	127	260	463	
	HDD18.3	5468	849	732	664	474	343	214	157	171	248	384	510	721	
	CDD10.0	276	0	0	0	0	6	49	103	89	27	2	0	0	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	6.7	7.9	7.6	7.2	6.4	5.4	5.7	6.1	6.0	6.2	6.7	7.4	7.7	
Precipitation	PrecAvg	1057	76	70	77	73	89	91	106	91	98	101	87	98	
	PrecMax	1201	137	117	140	135	150	175	219	168	266	168	171	147	
	PrecMin	817	26	33	34	30	38	21	32	36	35	30	33	48	
	PrecStd	91	29	22	30	30	29	38	38	33	43	35	28	24	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	4.1	7.2	11.2	16.9	20.1	21.6	20.7	18.4	14.1	11.2	5.2	
		MCWB	4.1	3.1	4.7	7.0	11.8	14.1	17.2	16.0	14.9	11.9	8.9	4.1	
	2%	DB	2.2	1.7	4.7	8.6	14.2	17.8	19.3	18.7	16.1	12.0	7.8	3.2	
		MCWB	1.3	0.5	2.5	5.2	10.1	13.2	15.8	15.4	13.6	10.0	6.3	2.2	
	5%	DB	0.6	0.4	3.1	7.1	12.4	16.2	17.9	17.3	14.6	10.5	6.3	2.1	
		MCWB	-0.2	-0.5	1.4	4.3	8.9	12.5	14.9	14.6	12.5	8.7	4.9	1.1	
	10%	DB	-1.2	-0.7	2.0	5.9	11.0	14.9	16.6	16.2	13.4	9.1	5.2	1.1	
		MCWB	-2.1	-1.5	0.6	3.5	8.1	11.8	14.3	13.9	11.6	7.4	3.9	0.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	3.6	4.0	7.7	12.8	16.0	18.2	17.5	15.9	12.2	9.9	4.1
			MCDB	5.3	4.3	5.1	10.1	14.5	18.7	20.0	19.1	17.5	13.3	11.4	4.8
2%		WB	1.6	0.8	2.5	5.6	10.6	14.2	16.6	16.0	14.1	10.2	6.6	2.2	
		MCDB	2.2	1.4	3.7	7.6	12.8	16.3	18.5	17.6	15.6	11.5	7.4	2.8	
5%		WB	-0.1	-0.2	1.5	4.7	9.4	13.1	15.5	15.1	12.9	8.8	5.0	1.2	
		MCDB	0.6	0.4	2.6	6.3	11.6	15.2	17.1	16.7	14.2	10.1	5.9	1.8	
10%		WB	-2.2	-1.4	0.6	3.8	8.3	12.1	14.6	14.2	11.8	7.6	4.1	0.1	
		MCDB	-1.3	-0.7	1.6	5.3	10.4	14.2	16.2	15.7	13.3	8.8	4.9	0.7	
Mean Daily Temperature Range		5% DB	MDBR	5.9	5.7	5.2	5.0	5.8	6.6	6.9	6.7	5.5	4.0	3.8	4.6
			MCDBR	8.1	5.7	5.3	6.8	8.5	8.7	8.7	8.3	7.5	6.2	5.4	5.4
	MCWBR		8.4	5.7	4.5	4.7	5.3	5.6	5.8	5.5	5.6	4.9	4.9	5.4	
	5% WB	MCDBR	8.3	5.7	5.2	6.1	7.8	8.0	8.1	7.7	7.3	6.0	5.0	5.2	
		MCWBR	8.3	5.6	4.6	4.6	5.5	6.0	6.2	5.8	6.0	5.1	4.7	5.5	
Clear-Sky Solar Irradiance	taub	0.265	0.275	0.307	0.351	0.365	0.385	0.409	0.391	0.365	0.329	0.297	0.269		
	taud	2.356	2.318	2.292	2.315	2.354	2.362	2.326	2.381	2.438	2.520	2.469	2.394		
	Ebn at Noon	815	888	908	898	896	876	848	849	838	819	771	761		
	Edh at Noon	68	88	108	118	118	118	121	110	94	74	62	58		
All-Sky Solar Radiation	RadAvg	1.26	2.19	3.33	4.39	5.16	5.50	5.43	4.65	3.58	2.10	1.14	0.85		
	RadStd	0.13	0.24	0.25	0.38	0.48	0.40	0.53	0.30	0.23	0.17	0.12	0.08		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page